



Project Description

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Setup of the Original Model as in @Agu2023 and running Monte Carlo simulation

We implement the dynamic discrete-choice demand model of @Agu2023. A consumer i each period t chooses $y_{it} \in \{0, 1, \dots, J\}$ where $y_{it} = 0$ is “no purchase” and $y_{it} = j > 0$ is buying brand j . The endogenous state $x_{it} = (\ell_{it}, d_{it})$ records the last brand purchased and the duration since the last purchase. Prices have a persistent component z_{it} and a transitory (promotion) component e_{it} . With i.i.d. Type-I Extreme Value shocks the per-period payoff absent the shock is

$$u_{\alpha_i}(y, x, p) = \begin{cases} \alpha_i(\ell) + \gamma h(\mu_i) - \beta^{dep}(\ell) d & y = 0, \\ \alpha_i(j) + \gamma h(\mu_i - p(j)) - \beta^{sc}(\ell, j) & y = j > 0. \end{cases}$$

The block below sets up the primitives, solves the consumer’s Bellman equation by VFI (used only for simulating data), simulates a panel, and estimates $\theta = (\gamma, \beta^{sc}, \beta^{dep})$ by the sufficient-statistic Conditional Maximum Likelihood (CML) estimator described in Sections 3–4 of the paper.

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\begin{table}[H]
\centering
\caption{Simulation and CML estimation summary.}
\label{tbl:cml-summary}
\begin{tabular}{ll}
\toprule
Statistic & Value \\
\midrule
Consumers & 30000 \\
Periods & 52 \\
Brands & 2 \\
Observed purchases & 563464 of 1560000 \\
Purchase rate & 36.1\% \\
Sufficient-statistic cells with  $e_3=e_4$  & 573 \\
Price-contrast std. dev. & 10.140
\end{tabular}
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Nonzero price-contrast share & 55.7\% \\
Alternating-history share & 16.6\% \\
Depreciation spell cells & 8557 \\
Depreciation duration cap  $d^*$  & 2 \\
Minimum  $T$  for depreciation CML & 5 \\
Constant-promotion spell restriction & True \\
Alpha FE sample std. dev. & 0.249, 0.453 \\
Optimizer converged? & Yes \\
Optimizer iterations & 15 \\
Monte Carlo enabled? & Yes \\
Monte Carlo replications & 10 \\
Monte Carlo consumers & 30000 \\
Monte Carlo periods & 52 \\
\bottomrule
\end{tabular}
\end{table}

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\begin{table}[H]
\centering
\caption{CML parameter estimates.}
\label{tbl:cml-estimates}
\begin{tabular}{lrrrrrrrr}
\toprule
Parameter & True & CML & Std. err. & z-stat & p-value & CI 2.5\% & CI 97.5\% & Bias \\
\midrule
 $\gamma$  & 1.035 & 1.643 & 0.850 & 1.932 & 0.053 & -0.024 & 3.309 & 0.608 \\
 $\tilde{\beta}_{1,2}$  & 1.050 & 1.149 & 0.129 & 8.942 & 0.000 & 0.897 & 1.401 & 0.099 \\
\bottomrule
\end{tabular}
\end{table}

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\begin{table}[H]

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\centering
\caption{CML depreciation estimates from no-purchase spell histories.}
\label{tbl:cml-depreciation}
\begin{tabular}{lrrrrrrrr}
\toprule
Parameter & True & CML & Std. err. & z-stat & p-value & CI 2.5\% & CI 97.5\% & Bias \\
\midrule
 $\beta^{\text{dep}}_1$  & 0.250 & 0.219 & 0.022 & 9.987 & 0.000 & 0.176 & 0.262 & -0.031 \\
 $\beta^{\text{dep}}_2$  & 0.250 & 0.185 & 0.169 & 1.093 & 0.274 & -0.147 & 0.516 & -0.065 \\
\bottomrule
\end{tabular}
\end{table}

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\begin{table}[H]
\centering
\caption{General CML fit and test statistics.}
\label{tbl:cml-fit}
\begin{tabular}{ll}
\toprule
Statistic & Value \\
\midrule
Log-likelihood & -183.302 \\
Null log-likelihood & -397.173 \\
Pseudo  $R^2$  & 0.538 \\
AIC & 370.603 \\
BIC & 379.305 \\
LR test statistic & 427.743 \\
LR test df & 2 \\
LR test p-value & 0.000 \\
\bottomrule
\end{tabular}
\end{table}

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\end{table}

\begin{table}[H]

\centering

\caption{Monte Carlo simulation and CML estimation summary across 10 replications.}

\label{tbl:mc-dgp-summary}

\begin{tabular}{ll}

\toprule

Statistic & Value \\\

\midrule

Replications & 10 \\\

Consumers per replication & 30000 \\\

Periods per replication & 52 \\\

Brands & 2 \\\

Mean observed purchases & 556890.3 of 1560000 \\\

Mean purchase rate & 35.7\% \\\

Mean sufficient-statistic cells with $e_3=e_4$ & 721.2 \\\

Mean price-contrast std. dev. & 10.171 \\\

Mean nonzero price-contrast share & 58.4\% \\\

Mean alternating-history share & 16.2\% \\\

Mean depreciation spell cells & 8379.0 \\\

Depreciation duration cap d^* & 2 \\\

Minimum TS for depreciation CML & 5 \\\

Constant-promotion spell restriction & True \\\

Mean alpha FE sample std. dev. & 0.313, 0.315 \\\

Purchase optimizer convergence share & 100.0\% \\\

Mean purchase optimizer iterations & 15.6 \\\

Depreciation estimator availability share & 100.0\% \\\

Depreciation optimizer convergence share & 100.0\% \\\

Mean depreciation optimizer iterations & 9.1 \\\

\bottomrule

\end{tabular}

\end{table}

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\begin{table}[H]
\centering
\caption{Monte Carlo performance summary across 10 replications with  $N=30000$  and  $T=52$ }
\label{tbl:mc-summary}
\begin{tabular}{lrrrrrrrrrrrr}
\toprule
Parameter & True & CML & Std. err. & z-stat & p-value & CI 2.5\% & CI 97.5\% & Bias & Re
\midrule
 $\gamma$  & 1.035 & 1.149 & 0.546 & 2.103 & 0.035 & 0.078 & 2.219 & 0.114 & 10 & 0.530 &
 $\tilde{\beta}_{1,2}$  & 1.050 & 1.099 & 0.136 & 8.103 & 0.000 & 0.833 & 1.365 & 0.049 &
 $\beta^{\text{dep}}_1$  & 0.250 & 0.238 & 0.015 & 16.100 & 0.000 & 0.209 & 0.267 & -
0.012 & 10 & 0.019 & 8379.000 \\
 $\beta^{\text{dep}}_2$  & 0.250 & 0.269 & 0.127 & 2.120 & 0.034 & 0.020 & 0.517 & 0.019 & 10 &
\bottomrule
\end{tabular}
\end{table}
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\noindent\emph{Note:} The first CML table uses purchase histories (k,j,k,j) and $(k,k,$
purchase spell histories from Section 3.3, imposing the simulated duration cap $d^*=2$ a

Empirical Application of the Original Model

Brand	week_num	Brand 1 unique prices	Brand 2 unique prices
0	1	12.11, 14.08, 14.66, 14.66, 16.37, 18.61, 18.9...	9.56, 9.56, 9.56, 9.56, 9.56, 9.56, 9.56
1	2	12.81, 14.44, 14.50, 19.32, 19.36, 19.39, 19.4...	9.56, 9.56, 9.56, 9.56, 9.56, 9.56, 9.56
2	3	14.46, 14.47, 14.58, 14.58, 16.25, 16.92, 19.3...	9.56, 9.56, 9.56, 9.56, 9.56, 9.56, 9.56
3	4	12.72, 14.29, 14.40, 14.40, 15.94, 19.03, 19.1...	9.56, 9.56, 9.56, 9.56, 9.56, 9.56, 9.56
4	5	12.89, 14.56, 14.56, 14.58, 15.56, 16.25, 17.0...	9.56, 9.56, 9.56, 9.56, 9.56, 9.56, 9.56
5	6	14.37, 15.34, 16.31, 19.19, 19.44, 19.69	9.56, 11.25, 11.49, 11.59, 12.94, 14.3
6	7	12.35, 14.08, 14.08, 14.45, 14.45, 15.58, 16.0...	9.56, 9.56, 9.56, 9.56, 9.56, 9.56, 9.56

Brand	week_num	Brand 1 unique prices	Brand 2 unique prices
7	8	10.77, 10.77, 10.78, 11.08, 11.17, 11.18, 11.1...	9.56, 10.05, 10.16, 10.17, 10.25, 10.3
8	9	14.74, 16.46, 19.83, 19.87, 19.89, 19.91	9.56, 9.56, 9.56, 9.56, 9.56, 9.56
9	10	13.01, 13.64, 14.67, 14.73, 16.37, 19.77, 19.8...	9.56, 11.63, 12.15, 12.96, 13.01, 14.6
10	11	12.55, 12.87, 13.21, 14.88, 16.19, 16.86, 17.0...	9.56, 9.56, 9.56, 9.56, 9.56, 9.56
11	12	12.13, 13.05, 13.84, 14.70, 15.62, 15.93, 16.55	9.56, 9.56, 9.56, 9.56, 9.56, 9.56
12	13	11.80, 12.51, 12.55, 12.97, 13.49, 13.82, 14.0...	9.56, 10.74, 11.05, 11.55, 11.92, 11.9
13	14	10.33, 10.61, 10.71, 10.78, 10.94, 11.09, 11.1...	9.56, 9.89, 9.98, 10.02, 10.13, 10.33,
14	15	11.97, 12.76, 13.72, 13.95, 15.14, 15.93, 16.5...	8.00, 8.00, 8.00, 8.00, 8.00, 8.00, 8.00
15	16	12.72, 14.73, 16.16, 16.45, 16.45, 19.04, 19.3...	9.56, 9.56, 9.56, 9.56, 9.56, 9.56, 9.56
16	17	13.50, 14.49, 14.71, 14.71, 16.13, 16.13, 16.4...	9.56, 9.56, 9.56, 9.56, 9.56, 9.56, 9.56
17	18	8.00, 8.00, 8.00, 8.00, 8.00, 8.00, 8.00, 8.00...	8.52, 8.52, 8.57, 8.61, 8.62, 8.78, 8.7
18	19	14.20, 14.55, 14.55, 15.74, 16.22, 18.83, 19.1...	9.56, 9.56, 9.56, 9.56, 9.56, 9.56, 9.56
19	20	14.21, 14.21, 14.62, 14.62, 15.76, 16.54, 18.8...	9.56, 9.56, 9.56, 9.56, 9.56, 9.56, 9.56
20	21	14.40, 14.70, 19.25, 19.49, 19.55, 19.84	9.56, 9.56, 9.56, 9.56, 9.56, 9.56, 9.56
21	22	14.06, 14.41, 15.38, 16.03, 18.57, 18.84, 18.9...	9.56, 9.56, 9.56, 9.56, 9.56, 9.56, 9.56
22	23	9.59, 9.59, 9.59, 9.59, 9.59, 9.59, 9.59, 9.59...	9.59, 9.59, 9.59, 9.59, 9.59, 9.59, 9.59

We merge weekly recommended retail prices, discount prices, and promotion states from the price file. The offered price is the discount price when a brand is on promotion and the recommended retail price otherwise.

	WeekNum	RRP_Brand_1	Discount_Brand_1	Promo_Brand_1	Brand_1_price	RRP_Brand_1
0	1	24.95	NaN	0	24.95	11.95
1	2	24.95	NaN	0	24.95	11.95
2	3	24.95	NaN	0	24.95	11.95
3	4	24.95	NaN	0	24.95	11.95
4	5	24.95	NaN	0	24.95	11.95

Table 3: Price and promotion file merge summary.

Statistic	Value
Price/promo workbook	Price_Promo.xlsx
Price/promo sheet	Sheet2
Price/promo weeks	52
Unmatched scanner weeks dropped	53

Table 4: Empirical scanner-data sample summary.

Statistic	Value
Households	42816
Weeks	52
Household-weeks	2226432
Observed purchases	301027
Purchase rate	13.5%
Brand 1 purchases	66446
Brand 2 purchases	234581
Rows with pre-choice last brand	1697631
Price source	Weekly RRP/discount offer price

Table 5: Weekly RRP, discount, and offered-price support by brand.

Brand	Weeks with prices	Promo weeks	Median RRP	Median discount	Median offered price	Min
1	52	10	24.950	9.277	24.950	
2	52	6	11.950	10.000	11.950	

Table 6: Empirical purchase-history support for the original CML estimator.

Sample	Future promo match	Cells	A histories	B histories	Price-contrast std. dev.	Min
Strict original CML	Yes	154	33	121	11.155	
Relaxed diagnostic	No	345	145	200	13.826	

Table 7: Empirical CML estimates using strict original future promotion-state matching.

Parameter	CML	Std. err.	z-stat	p-value	CI 2.5%	CI 97.5%
γ	0.193	0.027	7.200	0.000	0.141	0.246
$\tilde{\beta}_{1,2}$	-0.067	0.300	-0.222	0.824	-0.655	0.522

Table 8: Relaxed empirical CML estimates without future price-state matching.

Parameter	CML	Std. err.	z-stat	p-value	CI 2.5%	CI 97.5%
γ	0.179	0.019	9.594	0.000	0.142	0.216
$\tilde{\beta}_{1,2}$	-0.036	0.168	-0.218	0.828	-0.365	0.292

Note: The relaxed empirical estimates use the same A/B purchase histories as the original estimator, but do not impose equality of the future weekly promotion vector. They should therefore be interpreted as a diagnostic empirical analogue rather than the exact sufficient-statistic CML estimator.